

Question Paper

Exam Date & Time: 22-Dec-2025 (02:00 PM - 05:00 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU

SEVENTH

SEMESTER B.TECH. DEGREE, MAKEUP EXAMINATIONS - DECEMBER 2025

Social Network Analysis [IT_4411]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) When we think about a single aggregate measure to summarize the distances between the nodes in a given graph, there are two natural quantities that come to mind. One is the diameter, which we define to be the maximum distance between any pair of nodes in the graph. Another is the average distance, which - as the term suggests - is the average distance over all pairs of nodes in the graph. In many graphs, these two quantities are close to each other in value. But there are graphs where they can be very different. Construct an example of a graph where the diameter is more than three times as large as the average distance. (2)
- A)

Course Outcome: "CO1" / Cognitive Level: "Application"

- B) A software company has mapped the internal communication network among its employees. The analysis shows: (4)
- **Riya**, the project manager, often connects developers from different teams who otherwise don't interact directly.
 - **Amit**, a senior developer, communicates quickly and frequently with almost everyone in the organization, making him central in the flow of information.
 - The company wants to identify key influencers and potential bottlenecks in information sharing to improve project coordination.
- (i) Analyze how Riya's high betweenness centrality could both help and hinder information flow within the company.
- (ii) Contrast the roles of Riya and Amit in terms of their influence on communication efficiency.

Course Outcome: "CO1" / Cognitive Level: "Analysis"

- C) Examine Breadth-First Search (BFS) algorithm to traverse the graph in the figure. 1c, starting from node 1. List the order in which nodes are visited. Show all the steps clearly. (4)

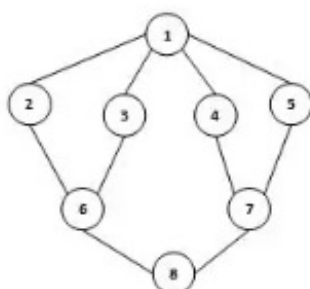


Figure 1c

- 2) Consider the social network represented in Figure 2a. Suppose that this social network was obtained by observing a group of people at a particular point in time and recording all their friendship relations. Now suppose that we come back at some point in the future and observe it again. According to the theories based on empirical studies of triadic closure in networks, Determine which new edge is most likely to be present? (5)
- A)

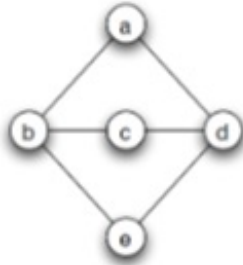
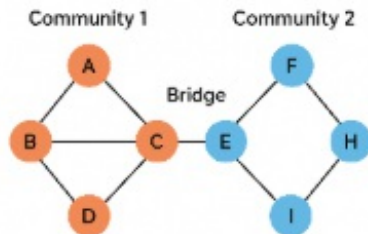


Figure 2a.

- B) A small online discussion forum is represented by the following social network graph, where each node represents a user, and an edge shows that two users frequently communicate. (5)



- Calculate the density of each community.
(Hint: $\text{Density} = 2 \times \text{number of edges} / n(n - 1)$, where n = number of nodes in the community)
- Evaluate whether the graph partition found gives a high modularity. Does it have many connections within communities and few between them?
- Assess the importance of node E using a betweenness centrality perspective.

- 3) Identify how triadic closure and social reinforcement contribute to clustering in friendship networks, explain Facebook friend graph example. (3)

A)

- B) Compare the implications of weak balance and strong balance for social tension reduction, using examples, Provide an example from a real-world platform. (3)

- C) Evaluate herding in online markets through the restaurant review example, How does Bayesian reasoning explain irrational popularity trends? (4)

- 4) Analyze the concept of network externalities and critical mass in the adoption of digital platforms. How do these concepts explain the dominance of a few network goods? (4)

A)

- B) Evaluate network effects and critical mass in social media platforms such as *Instagram* or (3)

WhatsApp, following the network goods analysis

Course Outcome: "CO3" / Cognitive Level: "Evaluate"

- C) Examine how clustered networks (Watts-Strogatz model) and random networks differ in cascade reach. How does clustering impede global diffusion? (3)

Course Outcome: "CO4" / Cognitive Level: "Analysis"

- 5) Evaluate how decentralized search operates in small-world networks. Discuss its implications for navigability and communication efficiency. (5)

A)

Course Outcome: "CO4" / Cognitive Level: "Evaluate"

- B) Examine similarity-based link prediction metrics like Common Neighbors, Jaccard Index, estimate future link formation. Discuss their strengths and limitations. (5)

Course Outcome: "CO4" / Cognitive Level: "Analysis"

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Event Name	Subject
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058-MIT BLR-MAKEUP EXAM-2025-2026 IT_4411:Social Network Analysis

#	Course Outcome	Description	Marks
1	1	CO 1	10
2	2	CO 2	13
3	3	CO 3	14
4	4	CO 4	13
Total			50

* **NA** - CO Not Selected

Event Name	Subject
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058-MIT BLR-MAKEUP EXAM-2025-2026 IT_4411:Social Network Analysis

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	0	0(0%)
3	Application	2	5(10%)
4	Analysis	6	23(46%)
5	Create	5	22(44%)
6	Evaluate	0	0(0%)
Total		13	50 (100%)